

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage:

50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

I **PAVAN JANA(192472313)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	
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1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s=255-r$ to the pixel values:

$r=[50,100,150,200]$

Compute the transformed values.

Grey Level Transformation

(a) Purpose of Grey Level Transformation in Image Enhancement

Problem Understanding (20%)

Grey level transformation is a point processing technique used in digital image processing. It modifies the intensity (brightness) values of pixels to improve the visual quality of an image or to highlight important features.

Method Selection

The required method is the **negative transformation**, which is defined by:

$$s=255-r \quad \text{or} \quad s=255-r$$

where:

- r = original pixel value
- s = transformed pixel value
- 255 = maximum intensity value for an 8-bit grayscale image

This transformation inverts the brightness levels of the image.

(b) Apply the Negative Transformation

Solution Process

Given pixel values:

$$r=[50, 100, 150, 200] \quad r = [50, 100, 150, 200] \quad r=[50,100,150,200]$$

Using the formula:

$$s=255-r \quad \text{or} \quad s=255-r$$

Calculate each value:

For $r=50$ $r = 50$ $r=50$:

- $s=255-50=205$ $s = 255 - 50 = 205$ $s=255-50=205$

For $r=100$ $r = 100$ $r=100$:

- $s=255-100=155$ $s = 255 - 100 = 155$ $s=255-100=155$

For $r=150$ $r = 150$ $r=150$:

- $s=255-150=105$ $s = 255 - 150 = 105$ $s=255-150=105$

For $r=200$ $r = 200$ $r=200$:

- $s=255-200=55$ $s = 255 - 200 = 55$ $s=255-200=55$

Accuracy of Calculation

Original Pixel (rrr) Transformed Pixel ($s=255-r$ $s = 255-r$ $s=255-r$)

50	205
100	155
150	105
200	55

Transformed Values:

$[205, 155, 105, 55]$ $\boxed{\{[205, \backslash; 155, \backslash; 105, \backslash; 55]\}}$ $[205, 155, 105, 55]$

Interpretation of Results

The negative transformation reverses the grayscale intensity values:

- Dark pixels become bright.
- Bright pixels become dark.
- This technique is useful for enhancing white or light details in dark regions, such as medical X-ray images or satellite imagery, where inverted contrast makes certain features easier to observe.

Final Answer

- **Purpose:** Grey level transformation enhances image quality by changing pixel intensity values to improve contrast, brightness, or highlight important features.
- **Transformed Pixel Values:** 205, 155, 105, 55 205, 155, 105, 55 205, 155, 105, 55

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s = c \log_{10}(1+r)$, where $c=10$, compute s for:

$r=[0,10,50,100]$

Log Transformation

(a) Purpose of Log Transformation

Log transformation is used to increase the brightness of dark (low-intensity) pixels while keeping bright pixels less affected. This helps reveal details in dark areas of an image

(b) Log Transformation

Formula:

$$s = 10 \log_{10}(1+r) \quad s = 10 \log(1+r) \quad s = 10 \log(1+r)$$

Given pixel values:

$$r = [0, 10, 50, 100] \quad r = [0, 10, 50, 100] \quad r = [0, 10, 50, 100]$$

Using **log base 10**:

Calculations:

- $r=0$
 $s = 10 \log_{10}(1+0) = 10 \log_{10}(1) = 10 \times 0 = 0$
 $s = 10 \log(1+0) = 10 \log(1) = 10 \times 0 = 0$
- $r=10$
 $s = 10 \log_{10}(11) \approx 10 \times 1.041 = 10.41$
 $s = 10 \log(11) \approx 10 \times 1.041 = 10.41$
- $r=50$
 $s = 10 \log_{10}(51) \approx 10 \times 1.708 = 17.08$
 $s = 10 \log(51) \approx 10 \times 1.708 = 17.08$
- $r=100$
 $s = 10 \log_{10}(101) \approx 10 \times 2.004 = 20.04$
 $s = 10 \log(101) \approx 10 \times 2.004 = 20.04$

Transformed values:

$[0, 10.41, 17.08, 20.04]$

Interpretation

Log transformation **makes dark pixels brighter** and **compresses bright pixel values**. This improves the visibility of details in dark regions of an image.

Final Answer

- **Purpose:** To enhance low-intensity (dark) pixels and improve image details.
- **Transformed values:** $[0, 10.41, 17.08, 20.04]$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity Frequency

0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

Histogram Processing

(a) Role of Histogram Equalization

Histogram equalization is used to **improve the contrast of an image**. It spreads the pixel intensity values over the available range, making dark areas brighter and bright areas clearer, so image details become easier to see.

(b) Histogram Equalization

Given Histogram:

Intensity Frequency

0	2
1	2
2	4
3	8

Step 1: Total Number of Pixels

$$2+2+4+8=16 \quad 2 + 2 + 4 + 8 = 16 \quad 2+2+4+8=16$$

Step 2: Compute Cumulative Distribution (CDF)

Intensity Frequency Cumulative Frequency CDF

0	2	2	$2/16=0.125$ $2/16 = 0.125$ $2/16=0.125$
1	2	4	$4/16=0.25$ $4/16 = 0.25$ $4/16=0.25$
2	4	8	$8/16=0.5$ $8/16 = 0.5$ $8/16=0.5$
3	8	16	$16/16=1.0$ $16/16 = 1.0$ $16/16=1.0$

Step 3: Compute Equalized Intensity Values

For a **3-bit image**, the number of intensity levels is:

$$L=2^3=8 \quad L = 2^3 = 8 \quad L=2^3=8$$

Formula:

$$s=(L-1) \times \text{CDF} = 7 \times \text{CDF} \quad s = (L-1) \times \text{CDF} = 7 \times \text{CDF} \quad s=(L-1) \times \text{CDF} = 7 \times \text{CDF}$$

Intensity CDF Equalized Value $7 \times \text{CDF}$ $7 \times \text{CDF}$ Rounded Value

0	0.125	0.875	1
1	0.25	1.75	2
2	0.50	3.50	4
3	1.00	7.00	7

Equalized Intensity Values:

$$[1, 2, 4, 7] \quad \boxed{[1, 2, 4, 7]} \quad [1, 2, 4, 7]$$

Interpretation

Histogram equalization **improves image contrast** by spreading the pixel values across the full intensity range. This makes the image appear clearer and reveals more details.

Final Answer

- **Purpose:** To improve image contrast by redistributing pixel intensities.

- **CDF: [0.125, 0.25, 0.50, 1.00]**
- **Equalized Intensity Values: [1, 2, 4, 7]**

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

Spatial Filtering (Smoothing)

(a) Purpose of Spatial Filtering for Smoothing

Spatial filtering is used to **reduce noise and smooth an image**. It replaces each pixel with the average of its neighboring pixels, making the image less noisy and smoother.

(b) Apply a 3 × 3 Averaging Filter

Given 3 × 3 Region:

$$\begin{matrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{matrix}$$

Step 1: Add all pixel values

$$10+20+30+20+30+40+30+40+50=270$$

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Step 2: Divide by the total number of pixels (9)

$$270 \div 9 = \frac{270}{9} = 30$$

New Center Pixel Value:

$$\boxed{30}$$

Interpretation

The averaging filter smooths the image by replacing the center pixel with the average of its neighboring pixels. This helps reduce noise and makes the image appear smoother.

Final Answer

- **Purpose:** To reduce noise and smooth the image.
- **New center pixel value: 30**

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

to the matrix:

$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 20 \\ 10 & 10 & 10 \end{bmatrix}$

Compute the output at the center.

Spatial Filtering (Sharpening)

(a) Purpose of Sharpening

Sharpening is a spatial filtering technique used to enhance edges and fine details in an image. It makes objects appear clearer by highlighting intensity changes between neighboring pixels.

(b) Apply Laplacian Mask

Given Laplacian Mask:

$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

Given Image Matrix:

$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$

Step 1: Multiply corresponding values and add

$$\begin{aligned} & (0 \times 10) + (-1 \times 10) + (0 \times 10) + (0 \times 10) + (-1 \times 10) + (0 \times 10) \\ & + (-1 \times 10) + (4 \times 20) + (-1 \times 10) + (-1 \times 10) + (4 \times 20) + (-1 \times 10) \\ & + (0 \times 10) + (-1 \times 10) + (0 \times 10) + (0 \times 10) + (-1 \times 10) + (0 \times 10) + (0 \times 10) + (-1 \times 10) + (0 \times 10) \end{aligned}$$

Step 2: Calculate

$$\begin{aligned} & = 0 - 10 + 0 - 10 + 80 - 10 + 0 - 10 + 0 = 0 - 10 + 0 - 10 + 80 - 10 + 0 - 10 + 0 \\ & = 0 - 10 + 0 - 10 + 80 - 10 + 0 - 10 + 0 = 40 = 40 \end{aligned}$$

Output at Center Pixel:

$\boxed{40}$

Interpretation

The Laplacian filter highlights the difference between the center pixel and its neighbors. A higher output value indicates the presence of an edge or detail in the image.

Final Answer

- Purpose: To enhance edges and image details.
- Output at center pixel: 40